

For this project our team developed a Clinical Decision Support (CDS) framework to support treatment selection for musculoskeletal patients on surgical or conservative treatment options. Using retrospective clinical data (from the first challenge/ Dataset A) we can estimate treatment effects across five key outcomes: long-term pain reduction, short-term treatment related pain, long-term functional improvement, rehabilitation time, and out of pocket cost. These treatment effects (TEs) capture how outcomes differ between surgery and conservative care, accounting for heterogeneity across patients.

In this stage, we put our model's results into practice, applying these TEs to a new batch of patients. In Stage 2, however, we are provided with the patients' preferences regarding what outcomes they would prefer to others, based upon a 100-point token system. To properly assess what treatment option would be best for a patient based upon these two factors, we converted treatment effects to a value scale using "willingness-to-pay" anchors for pain and functional outcomes, then penalty scaling for rehab time and out of pocket cost. The weighted priorities (WP) of the patients were first normalized and combined with the standardized TE results to calculate a final composite score for each patient.

The resulting scoring structure produced individualized treatment recommendations, where patients with positive scores ($\text{Score} > 0$) were recommended surgery, and those with negative scores were recommended to follow more conservative approaches. These results were then outputted to a prediction file, connecting the patient's id to their respective outcome.

This approach is intentionally designed to provide a deterministic recommendation to the patient and health professional involved. In practice, this approach would be implemented into a CDS tool that supports shared decision-making between clinicians and the patients. For one of our deliverables, we provide an example dashboard of what the output of one of these systems may look like, showing the breakdown of how the output variables (such as long-term function) stack up regarding a patient's preferences and recommended treatment.

By conducting this project, we have been able to provide a working example of how technological advancements in machine learning models, in conjunction with individual patient history and preferences, can be used to deliver quantifiable, useful results in the clinical world.